# Recursion and induction

## Programming · University · Lecture 2



# In this lecture

- Explain the call stack.
- Argue termination.
- Prove by induction.

# Remember

- One frame per call.
- A reachable base case.
- Induction mirrors code.

# Be the stack

1. Warm-up (5 min): Why did it crash?
2. Act (15 min): Student frames.
3. Trace (15 min): On paper.
4. Prove (20 min): Induction.
5. Wrap-up (5 min): One-minute paper.

# Let's discuss

1. What does each frame remember?
2. Why is fib slow?
3. What may you assume?

# Exit ticket

- Write the inductive hypothesis for a recursive list sum.